

Q1. Suppose you can enhance a machine either make multiply instructions run four times faster, or make memory access instructions run two times faster than before. You repeatedly run a program that takes 100 seconds to execute out of which 20% is used for multiplication, 50% for memory access instruction, and 30% for other tasks. What will be the speedup if you improve only multiplication? What will be the speedup if both improvements are made? [2+3]

Ans1. Given

Total time to execute = 100s

time used for Multiplication (Mul) = 20% of 100 = 20s

Time used for memory access instruction (Mem) = 50% of 100 = 50s

time used for other tasks (Oth) = 30% of 100 = 30s

(i) We improve only multiplication, i.e. we can make it 4 times faster.

$$\text{Mul} = \frac{20}{4} = \underline{5s} \text{ (4 times faster)}$$

Total time taken to execute the program

$$= 5 + 50 + 30 = 85s$$

$$\text{Speed up} = \frac{\text{original time to execute}}{\text{Time to execute with one modification}} = \frac{100}{85} \approx \underline{1.176}$$

(ii) We also improve memory access time

$$\text{Mem} = \frac{50}{2} = 25s \text{ (2 times faster)}$$

Total time to execute the program

$$= 5 + 25 + 30 = 60s$$

$$\text{Speed up} = \frac{\text{Original time to execute}}{\text{Execution time with all modification}} = \frac{100}{60} \approx \underline{1.67}$$

Q2. Consider the following fragment of C code: for ($i = 0$; $i < 100$; $i++$) $A[i] = B[i] + c$;
 Assume that A and B are array of words and the base address of A is in \$a0 and base address of B is in \$a1.
 Register \$t0 is associated with variable i and register \$s0 with c. Write the code for MIPS. How many instructions are executed during the running of this code? How many data references will be made during execution? [Code should be adequately commented] [4+0.5+0.5]

```

addi $t6, $zero, 100 # the loop termination
                        value
add $t0, $zero, $zero # i = 0
addi $t2, $a0, 0      # ptr to current
                        A[i]
addi $t3, $a1, 0      # ptr to current
                        B[i]
loop: lw $t4, 0($t3)   # load B[i]
add $t4, $t4, $s0      # B[i] + c
sw $t4, 0($t2)         # store in A[i]
addi $t0, $t0, 1       # i++
addi $t2, $t2, 4       # ptr to next
                        A[i]
addi $t3, $t3, 4       # ptr to next
                        B[i]
bne $t0, $t6, loop     # if i < 100,
                        goto loop.
    
```

The loop is executed 100 times and the loop contains 7 statements. There are 4 statements outside the loop. i.e., the

∴ the total number of instructions executed is $4 + (7 \times 100) = 704$

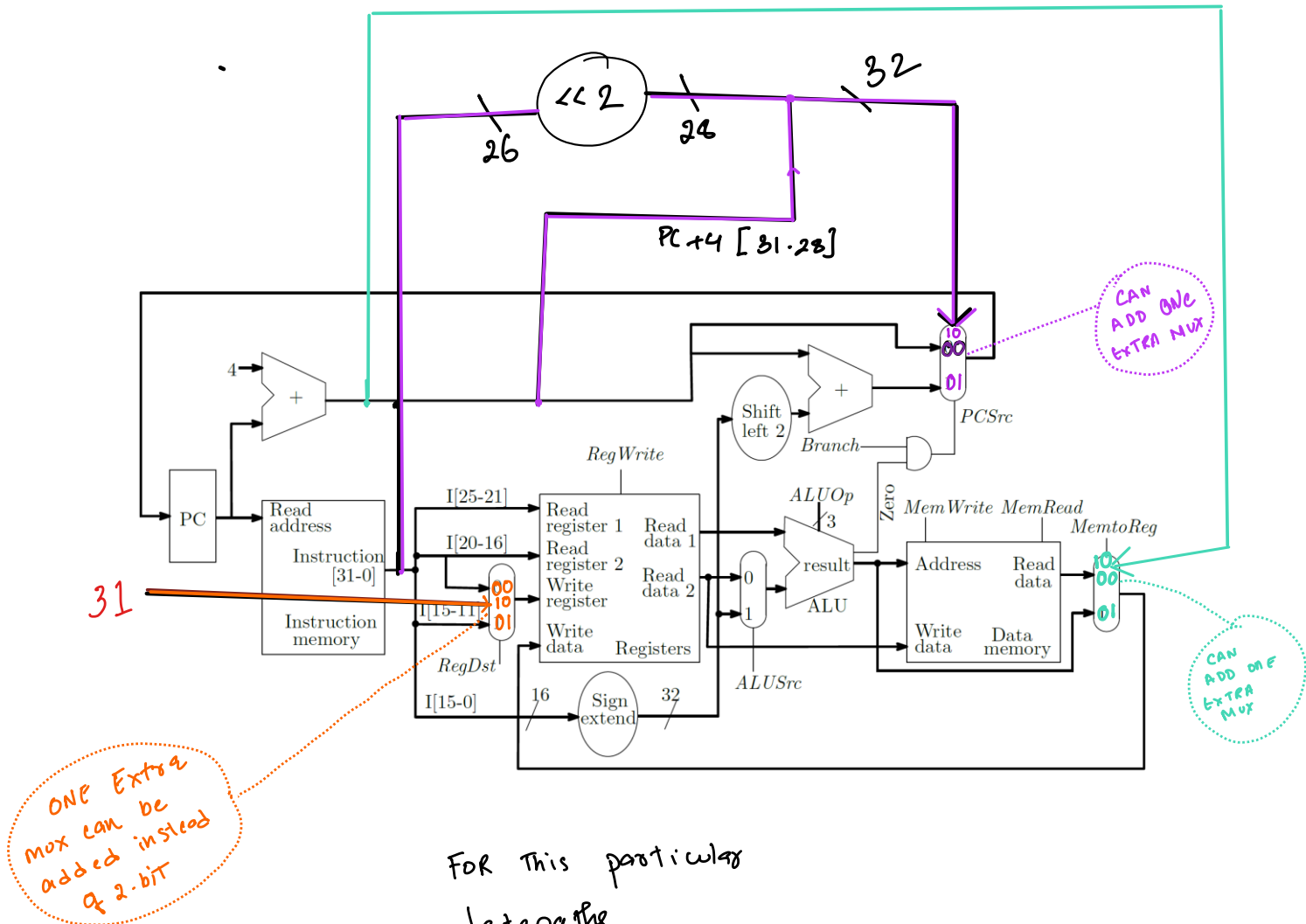
The total number of memory data reference: $100 \times 2 = 200$

Q3. We wish to add the instruction jal (jump and link) to the single cycle datapath. Add any necessary datapaths and control signals to the given datapath. Also provide each control signal values in your modified datapaths for executing jal. Instruction format of jal instruction is given below. (Note: \$ra = 31) [5]

#bits:	6	26
jal 10000	op	2500

$\$ra = PC + 4$, goto Target Address

Target address = $(PC + 4)[31:28]:(2500 \times 4)$



For this particular datapath,
Control signals for JAL is:

RegDst 10
PCSrc 10
MemtoReg 10

NOTE:

In case multiple Mux are added, the corresponding control signals need to be added